

Hanzla Sibghat

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Experience

Siemens AG

Feb 2025 – Present

Software Engineer (Working Student) – Munich, Germany

- Developed a full-stack enterprise Chrome extension using **React**, **TypeScript**, and **Node.js**, supporting AI-powered search, chat, and document analysis for thousands of internal users.
- Built end-to-end **RAG** pipelines integrating **LangChain**, **Ollama**, **GPT-4**, and **Mistral** with vector embeddings and **OCR/VLM**-based document processing.
- Architected **REST APIs** and async job pipelines on the backend while implementing rich frontend interfaces with **Redux**, owning features end-to-end from schema to UI.
- Containerized services with **Docker** and **CI/CD**, reducing deployment time from 30 minutes to under 5; achieved **80%+** test coverage with **Jest**.

National Database and Registration Authority (NADRA)

Feb 2024 – Sept 2024

Software Engineer – Islamabad, Pakistan

- Built end-to-end full-stack applications using **Java/Spring Boot** for backend and **Next.js** for frontend, supporting national identity workflows on **IoT** kiosks deployed nationwide.
- Designed **RESTful APIs** handling **10K+ requests per minute** while implementing complex multi-step UI flows with **Redux** state management and **Material-UI**.
- Optimized database performance through indexing, query rewriting, and caching, reducing average query latency by **70%**.
- Implemented secure communication between IoT kiosks and backend services with **TLS** encryption and signed payloads for tamper-proof data transmission.
- Wrote unit and integration tests across the stack using **JUnit**, **Mockito**, and **Jest**, contributing to a stronger code quality culture.

PLC Group

July 2023 – Feb 2024

Software Engineer – Ontario, Canada (Remote)

- Developed end-to-end features using **React** with **TypeScript** and **Node.js** with **GraphQL APIs**, supporting **100K+ concurrent users** with **100% zero-downtime** deployments.
- Designed efficient **GraphQL** schemas with **DataLoader**-based batching to eliminate N+1 queries, improving API response times by **60%**.
- Architected **Redis** caching layers for sessions and frequently accessed data, achieving **70%** reduction in average query latency.
- Implemented **React Query** for server-state and **Redux Toolkit** for client-state, creating predictable, performant data flows across complex UI hierarchies.
- Built test suites with **Jest** and **React Testing Library** for frontend and integration tests for backend, maintaining high coverage on critical flows.

Skills

Languages: TypeScript, JavaScript, Java, Python, SQL, C++

Frontend: React.js, Next.js, Redux, Redux Toolkit, React Query, Tailwind CSS, Material-UI, Shadcn/UI

Backend: Spring Boot, Node.js, Express.js, FastAPI, REST, GraphQL

Databases: PostgreSQL, MongoDB, MySQL, Redis, Oracle

Cloud & DevOps: Docker, AWS, CI/CD, GitHub Actions, Vercel

Testing & Tools: Jest, React Testing Library, Playwright, Cypress, JUnit, Git, Postman, Linux

Education

University of Passau

Oct 2024 – Present

MSc. in Computer Science – Passau, Germany

- Pursuing an MSc in Computer Science with focus on **Distributed Systems**, **Cloud Computing**, **Software Architecture**, and **Advanced Algorithms**, blending theoretical and practical expertise.

National University of Computer And Emerging Sciences (FAST-NUCES) Aug 2019 – Jun 2023
BSc. in Software Engineering – Islamabad, Pakistan

- Strong academic performance covering **full-stack development**, **data structures**, **algorithms**, **system design**, and **software architecture**, with a final-year capstone in IoT-based health monitoring.